Context-Aware Hierarchical Fusion for Drug Relational Learning

Yijingxiu Lu, Yinhua Piao, Sangseon Lee, and Sun Kim

Abstract—**The simultaneous use of multiple medications is a common practice in disease treatment, yet the same drug combination can lead to different effects under varying physiological, pharmacological, or genomic conditions—collectively referred to as the 'context'. Accurately predicting the outcomes of drug combinations across diverse contexts, also known as drug relational learning (DRL), is essential for improving therapeutic efficacy and safety. Despite its importance, existing methods face two major challenges: they are often tailored to specific DRL tasks, lacking generalizability, and they fail to explicitly model the influence of context on drug interactions. This limitation arises because most methods focus primarily on whole-drug compound structures, overlooking the fine-grained atomic-level interactions critical for context-aware predictions. To address these challenges, we propose a novel context-aware hierarchical fusion architecture for DRL. By formulating the problem as the label prediction of drug-drug-context triplets, our approach explicitly models the interaction between drugs by first learning their intrinsic atomic-level interactions and then incorporating context into their embeddings at the atomic level through information fusion. Experiments across diverse tasks—such as synergy prediction, polypharmacy side effect detection, and drug-drug interaction prediction—demonstrate our model's capability to effectively capture context-aware information. Importantly, our method consistently achieves robust performance in highly complex scenarios, highlighting its adaptability and utility in advancing context-aware drug relational learning.**

Index Terms—**Drug relational learning, hierarchical fusion, triplet relation learning.**

I. Introduction

THE simultaneous use of multiple medications, often referred to as polypharmacy, is a common practice in the treatment of various diseases [1]. However, this approach can lead to unpredictable drug reactions or interactions, potentially causing side effects, adverse drug events (ADEs), or antagonistic effects [2]. For instance, cabazitaxel and zoledronic acid exhibit synergistic effects in lung cancer cell lines but act antagonistically in breast cancer treatment. Predicting the outcomes of such medication combinations under varying molecular, physiological, pharmacological, or genomic conditions—collectively referred to as 'context'—has become a critical research focus [3], [4], [5]. Despite the significance of context in determining drug interaction, existing models often overlook its dynamic nature, instead assuming static relationships between drug pairs [6]. However, drug effects can vary significantly under different biological conditions, requiring a more adaptive and context-aware modeling approach. This emerging field, known as drug relational learning (DRL), seeks to address significant challenges in ensuring patient safety and optimizing treatment efficacy [7]. DRL encompasses various tasks, including: (1) Drug-drug interactions: chemical and physical interactions between drugs that can disrupt their intended functionality [8], (2) adverse drug reactions: unintended side effects arising from inherent pharmacological properties of drugs, which are often unpredictable during development and contribute significantly to morbidity and mortality [9], and (3) synergistic drug combinations: evaluate whether combination therapies achieve synergistic effects to enhance therapeutic efficacy for individual patients, a critical goal in precision medicine [10].

Exploring DRL presents numerous challenges. Traditional laboratory approaches to identify drug relationships are constrained by significant limitations, including high costs, labor intensive procedures, and long timelines [11]. The vast combinatorial space of potential drug interactions makes high-throughput screening and in vitro experiments impractical to systematically explore drug combination landscapes [12]. Moreover, clinical investigations of drug combinations not only carry the risk of exposing patients to unnecessary or harmful treatments, but also fail to adequately explore the combinatorial drug space during the preclinical stages [13]. These challenges highlight the pressing need for efficient and alternative strategies to comprehensively investigate drug relations. The development of computational methods, particularly machine learning, has enabled exploration of the complex and expansive space of drug combination relationships. Numerous computational approaches have been applied to drug relation learning and prediction tasks, making significant progress in the field [14], [15], [16]. Traditional machine learning methods often rely on the

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assumption of drug similarity, which posits that drugs with similar chemical structures or biological functions are likely to exhibit similar side effects [17]. These methods typically use molecular fingerprints or hand-crafted features to represent drugs, predicting relationships between drug pairs based on their chemical structural similarity [18]. However, such approaches are inherently limited by their reliance on predefined feature spaces, making them less effective when applied to novel or structurally diverse drug data. Recent advancements in deep learning have driven a shift toward graph-based learning methods, which automatically extract chemical structure information from raw molecular graph data. These methods have demonstrated notable success in predicting drug-drug interactions and other related tasks [19], [20], [21], [22]. By leveraging deep learning's ability to capture intricate molecular relationships, graph-based approaches offer a more flexible and generalizable framework for drug relational learning.

While structural similarity methods provide a foundation for drug relation prediction, two drugs with similar chemical structures may exhibit distinct biological activities or pharmacological properties. As a result, these methods often extract features that are not directly relevant to the specific prediction task [23], [24]. Furthermore, drug-drug interactions frequently arise from intricate biochemical mechanisms that cannot be fully explained by the properties of individual molecules. Many existing approaches focus predominantly on drug-level information, often neglecting the complex, mutual interactions between drug pairs—particularly those occurring at the atomic level [25]. Moreover, some methods typically assume that drug interactions are fixed and independent of external factors, whereas real-world drug effects are highly context-dependent [19], [26]. Contextual factors, such as cellular environments, patient-specific characteristics, or co-administered compounds, can significantly influence drug interactions, making it essential to incorporate contextual information into predictive models. Accurately modeling these mutual interactions within specific biological, chemical, or medical contexts requires a deeper understanding of how drugs interact with one another and their shared contextual semantics [27]. These fine-grained interactions are essential for uncovering the molecular features that drive observed effects, such as drug synergy or adverse reactions, under varying conditions. However, current research on modeling these atom-level mechanisms in a context-aware manner remains limited, highlighting a critical gap in the field.

Inspired by recent advancements in multimodal fusion [28], [29], [30], we propose a context-aware hierarchical cross-fusion model for predicting relationships between drug pairs. Following the transitivity rule in knowledge graphs, our model processes drug-drug-context triplets as input, sequentially learning drug-drug interactions and drug-context dependencies. Unlike conventional approaches that treat drug and context information at the same level, our hierarchical structure explicitly distinguishes drug-drug interactions from drug-context dependencies. This design ensures that drug relation representations are dynamically conditioned on contextual information rather than being statically encoded, thereby capturing the varying effects of drugs across different biological conditions. Furthermore, we introduce a novel atomic-level cross-fusion module that enables the model to selectively extract and integrate substructural interactions between drugs in a context-aware manner. By dynamically incorporating contextual influences, this mechanism ensures that the learned drug relation representations are not only reflective of their molecular properties but also finely tuned to the specific conditions governing their interactions. Experimental results across multiple drug relation datasets, covering both classification and regression tasks, demonstrate that our model consistently outperforms state-of-the-art baselines. Notably, as the influence of contextual factors increases, our model exhibits even greater performance improvements, underscoring its ability to effectively integrate and leverage contextual information beyond conventional approaches. Additionally, extensive ablation studies validate the soundness of our design, highlighting its effectiveness in addressing the complex challenges of drug relational learning.

II. Related Works

Drug relational learning plays a crucial role in various aspects of combination therapy research. Instead of monotherapy, combination therapy is widely employed in the treatment of cancer, viral diseases, and fungal infections due to its various advantages, including high efficacy, fewer side effects, and lower required dosages [3], [4], [5]. Identifying synergistic drug pairs for specific cancer types is thus crucial for improving the effectiveness of anticancer treatments. Additionally, predicting adverse events (AE) or adverse drug reactions (ADR) is another significant application of drug relational learning. AE and ADR are unexpected effects occurring at normal drug doses, which could be unpredictable during the drug development process [31]. Previous works heavily rely on quantitative structure-activity relationship (QSAR) for ADR prediction, which may fail in complex scenarios and lack generalization to new drugs [32], [33]. Developing a model that can uncover potential ADRs, even for new drug pairs, is needed. Another application of drug relational learning is drug-drug interaction, which could lead to a chain of undesired reactions, diminishing drug efficacy and even giving rise to hazardous side effects [34]. Identifying possible reactions that a pair of drugs could lead to is also within the scope of drug relational learning in DDI tasks.

Existing drug relational learning methods can be naturally categorized into two groups based on whether they utilize context information. Models that incorporate context information mostly leverage gene expression data from cancer cell lines to represent diseases [11], [27], [35]. However, these methods exhibit two main limitations: (1) Although they consider context information, most simply concatenate gene expression data with drug features, lacking a comprehensive understanding of the relationships between the two. (2) While these methods show significant efficacy in predicting drug synergy, their applicability to tasks such as DDI or ADR prediction is limited. On the other hand, the second group of methods that do not use context information, while achieving good performance in predicting a single type of side effect or interaction [19], [26], struggles to handle the large and complex space of context.

Drug relational learning methods can also be categorized into network-based and network-free, based on whether they utilize heterogeneous networks. Network-based methods construct network of drugs and biomedical concepts such as proteins and diseases to predict synergistic effects of co-administered drugs [36], [37] or multi-drug side effects [33], [38]. These methods leverage shared information from biomedical networks, demonstrating strong performance in drug relationship learning tasks. Such methods often require a large amount of interaction data, and they often struggle to extend to relationship prediction between unseen drugs in the network. Network-free approaches leveraging deep learning methods have emerged recently [11], [19]. The majority of them typically employ pre-defined drug features, such as extended connectivity fingerprints (ECFP) [39], or utilize various chemical and physical properties generated through Python packages like ChemoPy [40]. Instead of directly learning task-specific features with models, they rely on professionally crafted features based on prior knowledge. However, this reliance on expert features may not be sensitive to novel drug features, limiting the adaptability and learning capabilities of the model. Additionally, drugs with similar chemical structures do not necessarily exhibit identical biological activities [41]. The utilization of task-specific features learned through models has been demonstrated to enhance the predictive capability of models [26]. Furthermore, several deep learning methods, including attention mechanisms [27] and hypergraphs [42], have been employed in drug relational learning tasks.

III. Motivation

Learning relationships between drug-drug and drug-context is such crucial for solving tasks of this nature, whereas, existing research has seldom focused on learning these 'relations' when predicting whether a drug combination results in specific toxicity, side effects, or exhibits synergy in specific cell lines. In the process of drug relational learning:

1) As drug interactions change across different contexts, how to extract the crucial features from drugs that are conditioned to the given context?
2) As it is a complex biochemical process, is there any simple but robust way to model the problem?
3) Drug and context as two heterogeneous entities, how to properly learn the relations between them?

These three questions remark on the main challenges in drug relational learning.

IV. Preliminary

To address these challenges, we build on relational learning principles to design a context-aware hierarchical architecture for drug relational learning. In this section, we discuss how we address these challenges by employing the principle of relational learning, leading to the design of a context-aware hierarchical drug relational learning architecture.

a) Extracting hidden features conditioned: In relational learning, the relationship R between two entities A and B forms the minimal unit of a knowledge graph [43]. Using the inherent features of two entities to calculate their relationship is the most intuitive approach. However, due to the global and long-range dependencies that could be exhibited between different entities, using inherent features could lead the model bias to pseudo-correlated features when making predictions. The introduction of hidden variables with the assumption that the entities solely depend on these hidden variables in a relation has been proven to be effective in enabling the learning of non-local dependencies [44], [45]. In drug relational learning, global and long-range dependencies may arise from biochemical pathways or systemic interactions that extend beyond individual molecules. We thus apply individual entity encoders for both drugs and context to abstract their non-local task-related hidden representation. For a drug d and a context c, we abstract their hidden representations h_d and h_c using individual encoders g_d and g_c:

$$h_d = g_d(d) \quad , \tag{1}$$

$$h_c = g_c(c) \quad . \tag{2}$$

Detailed descriptions are in Section VI-A.

b) Learning relationships explicitly: Existing methods often rely on concatenating drug and context representations and feeding them into MLPs, which implicitly learn relations without explicit modeling. One of the main concerns of learning relations like this is the modeling of the relation of drug-drug-context triplet will become vague and less interpretable. To solve this problem, we propose to model the relations of drug-drug and the relations of drug-context independently and explicitly:

$$R_{d_i,d_j} = f_1(h_{d_i} h_{d_j}) \quad , \tag{3}$$

$$R_{d,c} = f_2(h_d, h_c) \quad . \tag{4}$$

Detailed descriptions are in Section VI-B.

c) Handling heterogeneous entities hierarchically: In relational learning, knowledge graphs often exhibit transitivity [46], a property that is also observed in drug relational learning. For instance, in DDI scenarios, if drug A alters the function of drug B, and this altered function of drug B leads to toxicity, then the co-administration of drugs A and B may result in toxicity. Similarly, in combination therapy for individual patients, drug synergy or side effects are conditioned on specific factors such as the cancer cell line or the patient's genomic profile. These examples illustrate that in drug relational learning, drugs can be seen as primary causal agents, while the context acts as a modulating factor or condition. This distinction further implies that the relationships between drugs and the context should be modeled sequentially, reflecting their hierarchical dependency.

Thus, we model the relation of a drug-drug-context triplet by fusing the interaction of drug pairs from drug to drug into the interaction of drug-context pairs from drug to context sequentially in a hierarchical architecture, detailed descriptions are in Section VI-B.

V. Problem Statement

We aim to learn context-aware representations of drug pairs for robust predictions in drug relational tasks. Here, 'context' refers to factors such as patient-specific conditions, pharmacological pathways, or molecular characteristics that influence the relations between drugs.

Let $D = \{d_1, d_2, \ldots, d_n\}$ denote a collection of n drugs, and $C = \{c_1, c_2, \ldots, c_m\}$ a set of m contexts. Each sample in dataset can be represented as a labeled triplet $(d_i, d_j, c_k; y)$, where $d_i, d_j \in D$, $c_k \in C$, and y is the target variable. For classification tasks, $y \in 0, 1$; for regression, $y \in \mathbb{R}$.

While some approaches treat drug relational learning as a multi-label classification problem, our framework instead models each (d_i, d_j, c_k) combination as an independent instance, enabling separate predictions for each annotation. This formulation provides flexibility in capturing context-specific drug relations and allows for effective generalization to unseen drugs and potentially unseen contexts.

To build a general model for context-aware learning of drug relations, we consider the three most popular tasks in disease treatment: *drug-drug synergy*, *drug-drug polypharmacy side effect*, and *drug-drug interaction* prediction tasks.

- *Drug-Drug Synergy:* predicts whether a drug pair d_i, d_j exhibit synergy under a specif cell line c_k.
- *Drug-Drug Polypharmacy Side Effect:* predicts whether a drug pair d_i, d_j causes an adverse event c_k.
- *Drug-Drug Interaction:* predicts whether a drug pair d_i, d_j induces a particular reaction c_k.

By treating each (d_i, d_j, c_k) as a distinct sample, our model learns to predict $P(Y|d_i, d_j, c_k)$ rather than aggregating multiple annotations into a single univariate representation. This ensures that the learned context embedding space effectively captures meaningful variations across different conditions.

VI. Methodology

In this work, inspired by multimodality fusion and the rule of transitivity in knowledge graph, we take drug-drug-context triplets as input and explicitly learn the relations of drug-drug and drug-context in a hierarchical architecture to ensure the learning of related drug latent representation conditioned with the other drug and a given context.

To achieve this, we initially develop uni-entity encoders to independently learn the hidden variables associated with drugs and contexts (Section VI-A). These hidden variables are subsequently employed to compute drug-drug and drug-context relations through a hierarchical cross-fusion process (Section VI-B). Ultimately, the learned representation is utilized for drug relational learning (Section VI-C).

A. Uni-Entity Encoder

1) Drug Encoder: Graph-based methods, known for their potent capability to learn topological information, have found extensive application in extracting features from drug data [19], [47]. In this study, we represent drugs using two-dimensional molecular graphs. Specifically, for each drug $d_i \in D$, we construct a molecular graph $G_i = (V, E)$ based on its SMILES string [48]. Here, $V = \{\mathbf{v}_1, \ldots, \mathbf{v}_N\}$ represents the set of atoms within the molecule, and $E \subseteq V \times V$ represents the chemical bonds connecting atoms. We calculate a node feature matrix $X \in \mathbb{R}^{N \times F}$ associated with intermolecular pharmacophores using RDKit [49].

Given a node v in the graph of a drug, we employ Graph Isomorphism Network (GIN) [50] to learn node representations h_v. We initialize each node representation using the $x_v \in \mathbb{R}^F$ in node feature matrix X as:

$$h_v^0 = x_v \tag{5}$$

We use $u \in \mathcal{N}(v)$ to denote the neighboring nodes of the node v in a drug 2D graph. At each layer l, we update node features according to the following equation:

$$h_v^{(l)} = \text{MLP}\left(h_v^{(l-1)} + \sum_{u \in \mathcal{N}(v)} h_u^{(l-1)}\right) \tag{6}$$

MLP denotes Multi-Layer perceptron, and $h_v^{(l)} \in \mathbb{R}^{F'}$ denotes the lth embedding vector associated with node v.

We set the total number of layers = 3 and utilize the final node representations for downstream information fusion.

2) Context Encoder: In different tasks, the term ‘context’ holds different real-world implications. In drug-drug synergy tasks, context represents the cell line for which a drug pair is intended to treat. Models trained specifically for predicting drug-drug synergy often use the gene expression profiles of cancer cell lines as context features. In drug-drug polypharmacy side effects and drug-drug interaction tasks, the context signifies a specific adverse event or reaction type, usually encoded as categorical features. Some models treat the task as a multi-label classification to facilitate learning.

As previously mentioned, in this study, we aim to explore factors influencing drug relational learning and to construct a general model. Therefore, for all three tasks, we utilize categorical feature encoding, specifically, one-hot encoding, as the input for context.

Given a context set $C = \{c_1, \ldots c_m\}$, we initialize the feature for each context $c_k \in C$ and use fully-connected layers to encode the latent context representation as follow:

$$H_k = \text{MLP}\left(x_{c_k}\right) \tag{7}$$

$x_{c_k} \in \mathbb{R}^M$ denotes the initialized one-hot encoding feature embedding of context c_k, and $H_k \in \mathbb{R}^{F'}$ represents the hidden embedding of context c_k.

B. Hierarchical Cross Fusion

In drug relational learning, existing methods often overlook the learning of context information or simply concatenate individually encoded drug and context features before feeding them into a predictor. In this study, we explicitly address the relation between context and drugs to achieve a deeper understanding of the target within drug-context pairs. Recognizing the heterogeneity between drugs and context entities, we introduce a hierarchical structure that sequentially updates homogeneous relations between drugs and heterogeneous relations between drugs and context (Fig. 2).

Specifically, considering that in reality, drug A can alter the function of drug B, and drug B can similarly alter the function

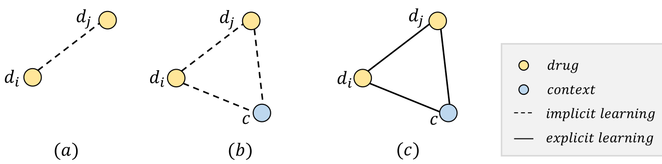

Fig. 1. Methods for drug relational learning. Current methods either (a) omit context information or (b) learn the pair-wise relations in drug-drug-context triplets implicitly. In this work, we (c) explicitly learn all pair-wise relations in each triplet to improve the robustness of prediction performance.

of drug A, the interaction between drugs exhibits asymmetry, we decompose the drug-drug relation R_{d_i,d_j} into $R_{d_i \to d_j}$ and $R_{d_j \to d_i}$. As for drug-context relations R_{d_i,c_k} and R_{d_j,c_k}, with drugs acting as causes and contexts as effects in biological, chemical, and pharmacological reactions, for practical significance, we only consider the two unidirectional relations from drugs to context, $R_{d_i \to c_k}$ and $R_{d_j \to c_k}$.

Therefore, in drug relational learning, we learn all pairwise relationships within triplets through the following two steps:

- *Drug-Drug Cross Fusion Step* In this step, we compute the relationships $\mathbf{R_{d_i \to d_j}}$ from drug i to drug j and vice versa. Its practical significance represents any relationship that may exist between a pair of drugs.
- *Drug-Context Cross Fusion Step* In this step, based on the relationships calculated in the previous step between pairs of drugs, we further compute the relationships from drugs to context to constrain the model's final learning content. Specifically, leveraging the transitivity of relationships, we extend the learned relationship from drug i to drug j to context k in this step. This extension helps extract the portions of the interaction between drug i and drug j that impact the context k, denoted as $R_{d_j \to c_k}$. The information from drug j to i is learned in a similar way.

1) Drug-Drug Cross Fusion Module: Inspired by the work of [51], we employ an atom-wise interaction map to calculate the directional relationship $R_{d_i \to d_j}$ between a pair of drugs i and j. Specifically, the interaction map is defined as $I_{ij} = \mathrm{sim}(H_i, H_j)$, where $H_i \in \mathbb{R}^{N_i \times F'}$ and $H_j \in \mathbb{R}^{N_j \times F'}$ represent the hidden representations of all nodes in drug i and drug j, and N_i and N_j represent the number of nodes in drug i and drug j. Consequently, we define $R_{d_i \to d_j} \in \mathbb{R}^{N_i \times F'}$ as:

$$R_{d_j \to d_i} = I_{ij}^T \cdot H_j \tag{8}$$

2) Drug-Context Cross Fusion Module: Based on the relationships computed in Section VI-B1, we update the representation $H_i' \in \mathbb{R}^{N_i \times 2F'}$ of drug i using the following equation:

$$H_i' = concat\left(H_i \parallel R_{d_j \to d_i}\right) \tag{9}$$

Similarly, we utilize the hidden representations of drugs and hidden representations of context from Section VI-A2 to compute the relations between drugs and context:

$$I_{i,k} = sim(H_i', H_k) \quad , \tag{10}$$

$$R_{d_i \to c_k} = I_{i,k}^T \cdot H_i' \quad . \tag{11}$$

sim denotes cosine similarity, $R_{d_i \to c_k} \in \mathbb{R}^{2F'}$ denotes the embedding of drug i fused by context k. Likewise, $R_{d_j \to c_k} \in \mathbb{R}^{2F'}$ denotes the embedding of drug j fused by context k.

C. Triplet Relation Predictor

We obtain the final hidden representation of the drug-drug-context triplet $h_{d_i,d_j,c_k} \in \mathbb{R}^{5F'}$, by combining the context hidden representation H_k with the hidden representation of relations that are calculated through the hierarchical cross-fusion module above:

$$h_{d_i,d_j,c_k} = concat\left(H_k \parallel R_{d_i \to c_k} \parallel R_{d_j \to c_k}\right) \tag{12}$$

By feeding the hidden representation h_{d_i,d_j,c_k} into a single-layer MLP, our model produces the final output $\hat{y}$, which represents either the probability of a positive label for classification tasks or a normalized regression score in [0,1]. The model is trained end-to-end by minimizing the binary cross-entropy (BCE) loss for the classification task:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left(y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right), \tag{13}$$

and mean squared error (MSE) loss for regression task:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2 . \tag{14}$$

All model parameters, including those of the drug and context encoders, are optimized jointly through backpropagation, ensuring that the learned representations are directly aligned with the final prediction objective.

VII. Experiments

A. Set-Up

1) Dataset: We use six popular benchmark datasets addressing three drug relational learning tasks: combined drug synergy, polypharmacy side effect, and drug-drug interaction. We use four classification tasks from chemicalx [7] and two regression tasks from TDC [34]. Details of datasets are summarized in Table I.

a) Four Classification Datasets from Chemicalx::

- *DrugComb* derived from wet-lab drug combination experiments, containing results from screening studies of drug combinations in various cancer cell lines. It contains 659,333 labels for the synergy between 4,146 drugs across 288 cancer cell lines [52], [53].
- *DrugCombDB* includes experimental data points from high-throughput screening assays of drug combinations, various sources including high-throughput screening and PubMed. It involves 191,391 labels for the synergy between 1,555 drugs across 112 human cell lines [54].
- *TwoSides* incorporates polypharmacy side effects for pairs of drugs, containing 225,070 significant associations between 644 drugs and 10 adverse events, with an equal number of negative samples that were generated without collisions of triples with positive labels [9].

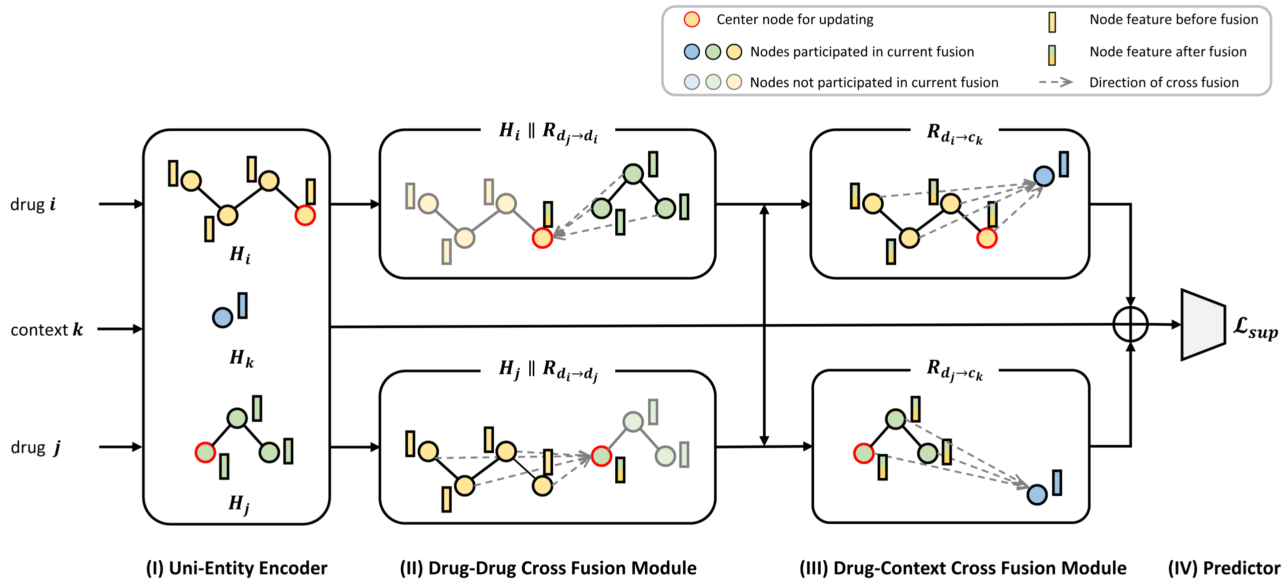

Fig. 2. Model architecture. Given a pair of drugs d_i and d_j and the context of interest c_k as input, our architecture predicts whether interaction exists in the certain triplet with hierarchical architecture. *Uni-Entity Encoder* learns the task-related hidden representation of drugs and context independently, after which *Drug-Drug Cross Fusion Module* calculates the possible relations between given drug pairs upon drug hidden representations. Considering drugs act as causes and context (e.g., cancer cell line) act as result in drug relational learning, we constrain the final relation representation with context information by fusing the drug-drug relations with a *Drug-Context Cross Fusion Module*.

TABLE I
Data Statistics

Task		Dataset	$\#D^1$	$\#D^2$	$\#C$	$\#DrugPair$	$\#Triplet$	Source
classification	Drug-Drug Synergy	DrugComb	4,117	895	288	74,378	659,333	chemicalx
		DrugCombDB	1,547	485	112	41,573	191,391	
	Drug-Drug Polypharmacy Side Effect	TWOSIDES	644	644	10	225,070	499,582	
	Drug-Drug Interaction	DrugbankDDI	1,706	1,706	86	365,056	383,496	
regression	Drug-Drug Synergy	OncoPolyPharmacology	37	37	39	583	23,052	TDC
		*DrugComb**	127	126	59	5,628	297,098	

$\#D^1$: number of drugs in the left. $\#D^2$: number of drugs in the right. $\#C$: number of context. #drugpair: number of drug-drug pairs. #triplet: number of drug-drug-context triplets.

TABLE II
Comparison of Baseline Models in Terms of Input Type, Backbone Architecture, and Multimodality Fusion Strategy

Model	Input		Backbone		Multimodality Fusion
	Drug	Context	Drug	Context	
GCN	2D graphs	One-Hot	GCN	MLP	Concatenation
GAT	2D graphs	One-Hot	GAT	MLP	Concatenation
DeepDrug	2D graphs	-	GCN	-	-
SSIDDI	2D graphs	-	GNN	-	-
CMRL	2D graphs	-	GIN	-	-
DeepSynergy	ECFP6, substructure	One-Hot	MLP	MLP	Concatenation
MatchMaker	chemical descriptor	One-Hot	MLP	MLP	Concatenation
DeepDDS	2D graphs	One-Hot	GCN	MLP	Concatenation
Ours	2D graphs	One-Hot	GIN	MLP	Cross-Fusion

- *DrugBankDDI* contains 86 DDI types, covering 192,284 DDIs from 1,706 drugs. Negative samples were generated without collisions of triples with positive labels [55].

b) Two Regression Datasets from TDC::

- *DrugComb** contains 297,098 drug synergy data across 59 NCI-60 cell lines with 129 drugs. It includes four numerical values representing various drug synergies, based on Bliss model, Highest Single Agent (HSA), Loewe additivity model, and Zero Interaction Potency (ZIP) model [52], [56].
- *OncoPolyPharmacology* contains 583 drug-drug combinations tested against 39 human cancer cell lines derived from seven tissue types. The synergy scores were calculated based on the Loewe additivity values [3], [11].

2) Baseline Models: To comprehensively evaluate our model, we compared it with (1) methods using different drug features and (2) methods with/without learning context information. Details of baselines are summarized in Table II.

a) Models categorized by types of input drug features: In terms of methods that use predefined drug features as input, *MatchMaker* utilized physicochemical features calculated by ChemoPy as drug inputs [35], [40]. *DeepSynergy* additionally employed extended connectivity fingerprints with a radius of 6 (ECFP6) and toxic substructure information [11], [57], [58]. Apart from the above models that use predefined drug features as input, *DeepDrug*, *SSIDDI*, *DeepDDS*, and *CMRL* utilized 2D molecular graphs to represent drugs. *DeepDrug* employed GCN, *SSIDDI* used GNN, *DeepDDS* employed GAT and GCN, while *CMRL* used GIN as the drug encoder [19], [26], [27], [47].

b) Models categorized by using or not using context information: Among the above models, only *DeepSynergy*, *MatchMaker* and *DeepDDS* utilize context information with drug features together as input. Specifically, *DeepSynergy* concatenates drug representations with context representation for prediction directly without using independent encoders. *MatchMaker* generates drug representation conditioned on context information for prediction. *DeepDDS* uses independent encoders to learn hidden representations of drugs and context representations for prediction.

3) Evaluation Protocol: Each experiment was repeated five times, with distinct random seeds used for initialization in each iteration. For the experiments involving DeepDrug, SSIDDI, DeepSynergy, MatchMaker, and DeepDDS under the random split setting, we used the results reported in [7], where the specific random seeds were not provided. For all other experiments, including those in the cold-drug setting, the experiments were conducted on an NVIDIA GeForce RTX 3090. We ensured that five distinct random seeds were applied consistently across all methods, in accordance with the baseline CMRL model [47], to facilitate fair comparisons. To ensure the comparability of the results between the aforementioned models and our methods, we re-evaluated DeepDrug, SSIDDI, DeepSynergy, MatchMaker, and DeepDDS under the random split setting using our five random seeds. The results were found to be consistent with those reported in [7], thus justifying the direct use of the reported results from that work.

For all the tasks, we use a 3-layer GIN [50] as drug encoder, a 2-layer MLP with latent dimension equal to 128 as context encoder, and a single layer of MLP without activation as a final predictor. The dataset was shuffled, and the model was initialized using each random seed. The experiments were repeated five times under each random seed to ensure a non-overlapping split of the dataset into training (80%), validation (10%), and test (10%) sets. Under each setting, our model was trained for 100 epochs with batch size 512, learning rate of 1×10^{-3}, and dropout rate 0.2. We apply Adam optimizer for model optimization, and set the learning rate decreasing on plateau by a factor of 10^{-1} following previous work [47].

For all baseline models, we adopted the best reported hyperparameter settings, including learning rate, number of training epochs, and optimization strategies from [7], [47]. During training, we employed an early stopping mechanism to optimize model selection. Specifically, we monitored the AUROC and accuracy scores on the validation set, terminating training if no improvement was observed over 20 consectutive evaluations. The final model selected for testing was the checkpoint that achieved the highest validation performance. Subsequently, we evaluated this model on the test set to obtain the final reported accuracy.

4) Evaluation Metrics: In classification tasks, we evaluate the performance of our model using AUROC, AUPRC, and F1. For regression tasks, we assess our model using root mean squared error (RMSE), Spearman's rank correlation coefficient (SCC), and Pearson's correlation coefficient (PCC).

B. Overall Performance

We have compiled the performance of our models on classification tasks in Table III, and the results for regression tasks are presented in Tables IV and V. Our models consistently outperform the baselines across all tasks. Meanwhile, the majority of baseline models show inconsistent performances across different datasets. For instance, DeepSynergy exhibits better performance on DrugComb and DrugBankDDI, while its performance on DrugCombDB is not good enough. Similarly, CMRL achieves satisfactory results on DrugCombDB and TwoSides, but its performance on DrugBankDDI is even lower by approximately 2.5% compared to GCN and GAT. These results underscore the effectiveness of our architecture in learning complex drug relations across diverse tasks. In this section, we discuss the advantages of our architecture over baselines by comparing them with several different criteria.

1) Importance of Learning Drug Features in Drug Relational Learning: Among all the baseline models, MatchMaker and DeepSynergy employ drug chemical descriptors calculated by the ChemoPy as inputs, while the remaining methods encode drug features using graph-based techniques. It is observed that the use of predefined drug descriptors can indeed lead to satisfactory model performance on specific datasets. For instance, both MatchMaker and DeepSynergy demonstrated competitive performance with other baselines on DrugBankDDI; however, on other datasets, their performance was less effective compared to graph-based representations.

To address why these methods perform exceptionally well on DrugBankDDI, we conducted a comparison of data statistics across different datasets, as shown in Table I. It is noteworthy that in the DrugBankDDI dataset, the number of drug-drug pairs is nearly equivalent to the number of drug-drug-context triples. We hypothesize that under such circumstances, where the model can distinguish most drug relations even without knowing context information, well-learned drug representations contribute more to prediction performance. The outstanding performance of our model underscores its ability to learn task-specific latent drug representations.

However, models using predefined features are not always the optimal choice, as they may even fall short of vanilla GNN models on certain datasets (DeepSynergy on DrugCombDB, MatchMaker on DrugComb and TwoSides). This emphasizes the significance of learning unbiased hidden features for drug relational learning. The consistently high performance of our model once again highlights the importance of extracting conditioned drug representations.

TABLE III
PERFORMANCE ON CHEMICALX DRUG RELATIONAL LEARNING TASKS (CLASSIFICATION)

	DrugComb			DrugCombDB			TwoSides			DrugBankDDI		
	AUROC ↑	AUPRC ↑	F1 ↑	AUROC ↑	AUPRC ↑	F1 ↑	AUROC ↑	AUPRC ↑	F1 ↑	AUROC ↑	AUPRC ↑	F1 ↑
GCN (w/)	66.1 (0.2)	72.3 (0.2)	70.5 (0.3)	76.3 (0.2)	60.7 (0.4)	47.3 (0.9)	92.6 (0.1)	90.8 (0.2)	86.1 (0.3)	97.1 (0.0)	96.0 (0.0)	92.5 (0.1)
GAT (w/)	66.2 (0.2)	72.4 (0.1)	70.7 (0.6)	76.1 (0.2)	60.5 (0.4)	47.1 (0.6)	92.4 (0.1)	90.6 (0.2)	85.6 (0.5)	97.0 (0.0)	95.8 (0.0)	92.5 (0.0)
DeepDrug	64.3 (0.1)	70.3 (0.2)	72.4 (0.1)	74.0 (0.1)	57.3 (0.1)	43.5 (1.0)	92.3 (0.4)	90.4 (0.2)	85.7 (0.2)	86.1 (0.3)	82.7 (0.3)	80.5 (0.2)
SSIDDI	66.3 (0.2)	72.6 (0.3)	70.4 (0.3)	77.9 (0.3)	62.4 (0.7)	52.8 (2.8)	91.7 (0.3)	89.8 (0.5)	83.8 (2.9)	85.5 (0.4)	82.2 (0.4)	79.0 (0.5)
CMRL	68.5 (0.2)	74.8 (0.1)	70.3 (0.0)	81.3 (0.1)	67.0 (0.3)	58.1 (0.8)	94.5 (0.0)	92.9 (0.2)	**91.7 (0.0)**	94.6 (0.2)	91.1 (0.0)	91.3 (0.6)
DeepSynergy (w/)	70.2 (0.3)	75.8 (0.3)	72.5 (0.2)	76.3 (0.5)	59.8 (0.8)	48.8 (1.2)	94.0 (0.1)	91.9 (0.1)	88.7 (0.1)	99.2 (0.1)	98.7 (0.1)	96.8 (0.1)
MatchMaker (w/)	66.2 (0.2)	72.5 (0.1)	71.2 (0.2)	78.6 (0.4)	63.7 (0.9)	50.8 (2.0)	91.2 (0.2)	89.2 (0.1)	84.9 (0.1)	98.7 (0.1)	98.1 (0.1)	95.9 (0.1)
DeepDDS (w/)	69.1 (0.2)	74.9 (0.2)	73.0 (0.3)	79.0 (0.6)	64.6 (1.0)	53.7 (1.3)	94.1 (0.0)	92.2 (0.0)	88.2 (0.2)	98.9 (0.0)	98.5 (0.1)	96.1 (0.1)
Ours	**81.8 (0.1)**	**85.7 (0.0)**	**77.8 (0.2)**	**86.0 (0.1)**	**74.8 (0.2)**	**66.4 (0.7)**	**95.6 (0.0)**	**93.4 (0.2)**	91.6 (0.1)	**99.8 (0.0)**	**99.6 (0.0)**	**99.0 (0.0)**

W/ indicates using context information.

TABLE IV
PERFORMANCE ON TDC DRUG-DRUG SYNERGY TASK: DRUGCOMB* (REGRESSION)

	ZIP			HSA			Bliss			Loewe		
	RMSE ↓	SCC ↑	PCC ↑	RMSE ↓	SCC ↑	PCC ↑	RMSE ↓	SCC ↑	PCC ↑	RMSE ↓	SCC ↑	PCC ↑
DeepDrug	5.35 (0.02)	0.38 (0.00)	0.34 (0.01)	5.88 (0.03)	0.31 (0.01)	0.32 (0.00)	6.02 (0.03)	0.26 (0.00)	0.29 (0.00)	17.18 (0.06)	0.06 (0.01)	0.11 (0.01)
SSIDDI	4.56 (0.02)	0.48 (0.00)	0.53 (0.00)	4.93 (0.02)	0.48 (0.00)	0.56 (0.01)	5.13 (0.03)	0.42 (0.01)	0.55 (0.00)	11.52 (0.03)	0.55 (0.00)	0.61 (0.00)
CMRL	4.26 (0.01)	0.49 (0.01)	0.54 (0.01)	4.59 (0.01)	0.50 (0.00)	0.58 (0.00)	4.78 (0.01)	0.45 (0.00)	0.57 (0.00)	11.23 (0.03)	0.57 (0.00)	0.63 (0.00)
DeepSynergy (w/)	4.02 (0.01)	0.61 (0.00)	0.66 (0.00)	4.38 (0.02)	0.57 (0.00)	0.68 (0.00)	4.55 (0.02)	0.52 (0.00)	0.67 (0.00)	10.63 (0.09)	0.61 (0.01)	0.69 (0.00)
MatchMaker (w/)	4.65 (0.02)	0.51 (0.00)	0.51 (0.00)	5.32 (0.03)	0.42 (0.00)	0.47 (0.01)	5.50 (0.03)	0.38 (0.01)	0.46 (0.01)	13.20 (0.31)	0.39 (0.04)	0.44 (0.06)
DeepDDS (w/)	4.56 (0.02)	0.51 (0.00)	0.54 (0.01)	5.17 (0.04)	0.45 (0.00)	0.53 (0.01)	5.40 (0.03)	0.40 (0.01)	0.51 (0.01)	12.01 (0.12)	0.50 (0.01)	0.58 (0.00)
Ours	**3.38 (0.03)**	**0.69 (0.00)**	**0.74 (0.00)**	**3.67 (0.06)**	**0.65 (0.01)**	**0.76 (0.01)**	**3.93 (0.03)**	**0.61 (0.01)**	**0.74 (0.00)**	**9.30 (0.08)**	**0.69 (0.00)**	**0.76 (0.00)**
* random split												
DeepDrug	5.29 (0.02)	0.08 (0.01)	0.12 (0.00)	5.96 (0.05)	0.08 (0.01)	0.10 (0.01)	6.04 (0.05)	0.03 (0.02)	0.06 (0.03)	14.27 (0.07)	0.02 (0.01)	0.02 (0.01)
SSIDDI	6.00 (0.34)	0.07 (0.04)	0.07 (0.06)	7.09 (0.50)	0.05 (0.02)	0.05 (0.01)	7.42 (0.41)	0.10 (0.05)	0.07 (0.04)	19.02 (0.44)	0.01 (0.01)	0.01 (0.00)
CMRL	5.36 (0.26)	0.15 (0.02)	0.16 (0.02)	5.29 (0.00)	0.16 (0.01)	**0.17 (0.01)**	5.81 (0.38)	0.12 (0.02)	0.12 (0.02)	15.31 (0.44)	0.03 (0.02)	0.03 (0.02)
DeepSynergy (w/)	6.02 (0.23)	0.14 (0.04)	0.15 (0.05)	6.72 (0.17)	0.10 (0.03)	0.10 (0.05)	6.76 (0.43)	0.07 (0.02)	0.06 (0.03)	15.28 (0.26)	0.06 (0.01)	0.06 (0.01)
MatchMaker (w/)	5.15 (0.02)	**0.22 (0.00)**	0.21 (0.00)	5.90 (0.11)	0.12 (0.02)	0.12 (0.01)	5.78 (0.02)	0.13 (0.01)	0.10 (0.01)	**13.76 (0.04)**	**0.14 (0.02)**	**0.16 (0.01)**
DeepDDS (w/)	5.12 (0.01)	0.19 (0.02)	0.21 (0.03)	**5.70 (0.05)**	0.09 (0.01)	0.09 (0.02)	5.82 (0.01)	0.05 (0.02)	0.07 (0.02)	14.59 (0.29)	0.05 (0.01)	0.04 (0.02)
Ours	**4.87 (0.11)**	0.21 (0.00)	**0.23 (0.00)**	5.27 (0.16)	**0.18 (0.02)**	0.18 (0.06)	**5.48 (0.12)**	**0.16 (0.04)**	**0.16 (0.05)**	16.34 (1.70)	0.05 (0.02)	0.05 (0.02)
* cold drug split												

W/ indicates using context information.

TABLE V
PERFORMANCE ON TDC DRUG-DRUG SYNERGY TASK: ONCOPOLYPHARMACOLOGY (REGRESSION)

	OncoPolyPharmacology		
	RMSE ↓	SCC ↑	PCC ↑
GCN (w/)	19.86 (1.02)	0.51 (0.01)	0.46 (0.02)
GAT (w/)	19.88 (1.04)	0.51 (0.01)	0.46 (0.02)
DeepDrug	22.82 (0.69)	0.44 (0.01)	0.34 (0.02)
SSIDDI	19.74 (0.76)	0.54 (0.01)	0.49 (0.02)
CMRL	19.91 (0.99)	0.51 (0.01)	0.45 (0.02)
DeepSynergy (w/)	17.59 (0.59)	0.65 (0.02)	0.63 (0.02)
MatchMaker (w/)	18.79 (0.76)	0.59 (0.01)	0.56 (0.02)
DeepDDS (w/)	19.75 (0.75)	0.51 (0.01)	0.49 (0.01)
Ours	**15.28 (0.17)**	**0.73 (0.01)**	**0.74 (0.01)**

W/ indicates using context information.

TABLE VI
ABLATION STUDY RESULTS ON DRUGCOMBDB DATABASE

RL	H	$d_i \rightarrow d_j \rightarrow c_k$	$d_j \rightarrow d_i \rightarrow c_k$	AUROC	AUPRC	F1
Implicit	-			76.4 (0.1)	60.7 (0.1)	48.8 (0.2)
Explicit	w/o	-		82.7 (0.2)	69.5 (0.4)	60.9 (0.4)
	w/	✓	-	84.5 (0.4)	72.6 (0.6)	64.7 (0.8)
	w/	-	✓	84.0 (0.3)	71.7 (0.8)	63.0 (1.4)
	w/	✓	✓	**86.0 (0.1)**	**74.8 (0.2)**	**66.4 (0.7)**

RL: relational learning; **H**: hierarchy; $d_i \rightarrow d_j \rightarrow c_k$ and $d_j \rightarrow d_i \rightarrow c_k$ stand for the direction of relation fusion (section vi-b).

TABLE VII
PERCENT GAIN OF PERFORMANCE OF OUR MODEL WHEN COMPARED WITH BEST PERFORMED BASELINE MODEL (AUROC)

Dataset	$\#Triplet : \#DrugPair$	AUROC $gain\%$
DrugBankDDI	1.05	0.6
TwoSides	2.22	1.1
DrugCombDB	4.60	4.7
DrugComb	8.86	11.6

2) Significance of Using Context Information: In this section, we discuss the significance of incorporating context information. Notably, among all the baseline models, DeepDrug, SSIDDI, and CMRL abstain from utilizing context information, while DeepSynergy, MatchMaker, and DeepDDS incorporate context information into their predictions. We observe that on datasets where the number of drug-drug-context triplets is comparable to the number of drug-drug pairs, such as in TwoSides and DrugBankDDI, models can achieve comparable performance even without considering context information. However, these models exhibit significant performance instability on datasets where drug relations are highly sensitive to different contexts. For instance, CMRL's performance on DrugBankDDI is approximately 2.5% lower than that of GCN and GAT. In contrast, our model demonstrates robust performance, even as datasets become more complex due to increased contextual variations, underscoring the importance of hierarchical learning of context representation in such tasks.

Particularly noteworthy are the datasets OncoPolyPharmacology and DrugComb*, where the ratio of $\#Triplets : \#DrugPair$ reaches 39 and 52, respectively. In these tasks, our model consistently outperforms the baseline models across

TABLE VIII
PERFORMANCE UNDER COLD-DRUG SETTING ON DRUGBANKDDI DATASET (CLASSIFICATION)

	DrugBankDDI		
	AUROC ↑	AUPRC ↑	F1 ↑
DeepDrug	67.2 (1.0)	66.6 (1.3)	55.4 (1.0)
SSIDDI	66.2 (0.8)	65.9 (1.3)	53.0 (4.5)
CMRL	94.0 (0.2)	90.3 (0.3)	90.0 (0.2)
DeepSynergy (w/)	90.8 (0.6)	90.6 (0.5)	73.1 (2.1)
MatchMaker (w/)	92.8 (0.1)	93.3 (0.6)	74.7 (2.5)
DeepDDS (w/)	93.4 (0.5)	94.1 (0.4)	85.3 (0.6)
HypergraphSynergy (w/)	91.8 (0.4)	92.6 (0.5)	85.7 (0.8)
Ours	**99.6 (0.0)**	**99.4 (0.1)**	**98.8 (0.1)**

W/ indicates using context information.

Fig. 3. t-SNE embeddings of latent representations for test sets across four datasets (DrugComb, DrugCombDB, TwoSides, and DrugBankDDI). The colors in the t-SNE plots represent context labels, illustrating the contextual grouping of drug relations. Our model demonstrates clear and consistent clustering aligned with distinct contexts, even in complex datasets. For instance, in the DrugComb dataset, samples of different cancer cell lines are well-separated. In contrast, baseline models such as DeepDrug and SSIDDI fail to capture this information, with embeddings scattered across the plot. Other models, like DeepSynergy and DeepDDS, exhibit limited clustering, succeeding only in simpler datasets (e.g., TwoSides). These results emphasize our model's superior capability to learn and distinguish context-conditioned drug relations.

various metrics, demonstrating its promising ability to extract context-aware drug relations.

3) Contribution of Explicitly Learning Drug Relations: We observed an interesting phenomenon: as the ratio of (d_i, d_j, c_k) to (d_i, d_j) increases across different datasets, the performance gain of our model compared to other baselines becomes more significant. As shown in Table VII, with the $\#Triplet : \#DrugPair$ ratio increasing from 1.05 to 8.86, the performance gain of our model compared to the highest-performing baseline model (based on AUROC ranking) increased from 0.6% to 11.6%. These results not only demonstrate the successful learning of drug relations by our model, particularly in learning drug relations under specific contexts but also affirm that our model can effectively capture complex drug relations in datasets with significant context influence, maintaining superior performance over other models.

One of the most noteworthy distinctions between our model and other baselines is that our model explicitly learns drug relations through the drug-drug-context triplet, utilizing a hierarchical architecture. To quantitatively analyze the extent to which explicit learning contributes to the overall performance, we modified our architecture by directly concatenating the hidden drug representations and context representation for prediction, ensuring that the relation is implicitly learned through the final MLP layer. The results are presented in the first line of Table VI. It is evident that the model's performance experiences a significant drop when relations are not explicitly modeled, underscoring the necessity of explicit learning in drug relational learning.

4) Significance of Hierarchical Cross Fusion: Another significant aspect of our model, distinguishing it from the baselines, is the hierarchical cross-fusion architecture. We posit that hierarchically integrating information—first from drug to drug and subsequently from drug to context—enhances the model's ability to learn context-constrained features of drug pairs. In tasks such as drug relational learning, where the co-administration of drugs is considered, drugs often function as causal factors, while contexts, such as cancer cell lines, serve as the conditions under which interactions manifest.

To assess the contribution of each model component to performance improvements, we analyze three key variations of relational learning: (i) implicit relational learning, where relational patterns are learned through a simple concatenation-based MLP prediction (similar to DeepSynergy); (ii) explicit relational learning without hierarchy, where the drug-drug and drug-context fusion modules operate independently and in parallel; and (iii) a single-directional relational learning with hierarchy, where the drug-drug and drug-context fusion modules are applied in sequential with either $d_i \to d_j \to c_k$ or $d_j \to d_i \to c_k$ is applied. The results of this ablation study are presented in Table VI. In scenario (i), relational dependencies are implicitly modeled by directly concatenating the hidden representations of drugs and context, followed by MLP-based prediction. This results in a performance drop of approximately 3.3% in AUROC compared to our full model, suggesting that hierarchical fusion effectively filters out irrelevant features and enhances predictive accuracy. In scenario (ii), where fusion is performed without hierarchical structuring, we observe an improvement over scenario (i), demonstrating the effectiveness of our cross-fusion modules in capturing relational dependencies. Furthermore, the superior performance of this scenario compared to baseline methods indicates that atomic-level fusion mechanisms provide more informative representations of drug interactions than conventional concatenation-based approaches, such as DeepSynergy. In scenario (iii), where fusion is performed in only one direction, we observe a decline in performance relative to the original bidirectional mechanism, indicating that single-directional cross-fusion is insufficient for optimal drug relational learning. This result highlights the importance of our bidirectional cross-fusion mechanism, which successfully captures distinct and complementary information from both drugs, leading to more accurate predictions.

5) Cold-Drug Settings: To assess the generalization ability of our model in predicting relationships between unknown drug pairs, we adopted a cold-drug setting by partitioning a small subset of drugs from the original dataset. These drugs, along with their corresponding relationships, formed the test set. The remaining drugs, along with their context-conditioned relationships, constituted the training and validation sets. We then compared the performance of our model against existing baselines in this cold-drug setting.

Tables IV and VIII present the experimental results under the cold-drug setting. Notably, our model outperformed other models by a significant margin on DrugBankDDI, while achieving performance comparable to the best baseline on DrugComb*. We observed that on DrugBankDDI, where a diverse set of drugs is present, most models employing graph-based encoders, including ours, CMRL, and DeepDDS, performed well. This could be attributed to the abundance of different drugs in DrugBankDDI, allowing graph-based models to better learn structural features relevant to relationships. Conversely, DrugComb* contains 5,628 distinct drug pairs but has 186,810 drug-drug-triplet combinations, indicating high context diversity in this database. In such a context-rich environment, the ability of models to learn contextual information is more critical for performance. We observed that, apart from our model, MatchMaker and DeepDDS, which also learn context information, achieved promising results.

C. Case Study

To investigate the effectiveness of our context-aware hierarchical architecture in explicitly learning context-conditioned drug relations, we visualized the t-SNE embeddings of the last latent layer before the output for the test sets of each dataset. As shown in Fig. 3, the context information can be discerned more clearly in the latent layers of our model, indicating that our model adequately learns and considers the context-conditioned drug-drug-relations during prediction, which also suggests the significance of model's ability to learn and distinguish different contexts in DRL tasks.

VIII. Conclusion

Drug relational learning (DRL) is critical for predicting the outcomes of drug combinations under diverse physiological, pharmacological, or genomic conditions, collectively referred to as the 'context'. This capability is essential for improving therapeutic efficacy and ensuring patient safety. However, existing approaches often lack generalizability and fail to explicitly model drug interactions conditioned on specific contexts, particularly at the atomic level. In this study, we address these limitations by introducing a novel context-aware hierarchical fusion model for DRL. By formulating DRL as the label prediction of drug-drug-context triplets, our method explicitly captures the intrinsic atomic-level interactions between drug pairs and integrates contextual information into their embeddings through a hierarchical fusion mechanism. This design ensures a comprehensive representation of both drug interactions and their contextual dependencies.

Experimental results across diverse tasks, including synergy prediction, polypharmacy side effect detection, and drug-drug interaction prediction, validate the effectiveness of our model. Our approach consistently achieves superior performance compared to state-of-the-art baselines, particularly in scenarios where contextual influence is significant. Furthermore, ablation studies confirm the architectural soundness and demonstrate the model's ability to effectively represent context-aware drug relations at a fine-grained level. Moreover, while our current implementation employs a one-hot encoding representation for the context, our approach is inherently flexible and can accommodate richer numerical annotations, such as learned embeddings from large-scale models. Incorporating such representations could further enhance the model's ability to capture nuanced drug-context interactions and improve predictive performance. Exploring these alternatives remains a promising direction for future research. In conclusion, our work provides a unified and generalizable framework for context-aware drug relational learning, offering robust solutions to key challenges in polypharmacy and drug interaction prediction. By explicitly modeling context at the atomic level, our approach advances the state of the art in DRL and holds potential for broader applications in drug development and precision medicine.

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